The Timing of Early-Life Shocks and Adolescent Mental Health

October 2026

Do the effects of an earthquake in early childhood persist?



Shock 27 February 2010 (27-F), magnitude 8.8

Exposed Children aged 6 months to 4 years

Effects of early-childhood shocks

Birth weight (Torche, 2011)

Cognitive and non-cognitive development (Gillmore, 2026)

Height and learning (Andrabi, Daniels, and Das, 2023)

Adult mental health (Persson and Rossin-Slater, 2018)

This paper

Mental health in adolescence, ages 14–18

▸ PGA map

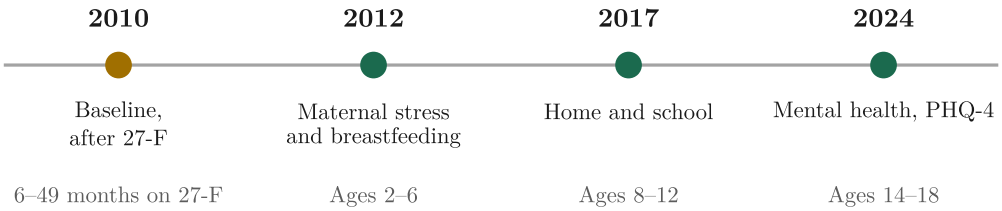
Does it matter at what age children live through the shock?

> "The effects of stress on the brain depend critically on the timing (age of onset and duration). When stress occurs early in life it can have profound and lasting effects on brain organization and function." (Gee and Casey, 2015, p. 184)

Potential mechanisms

	Channels
Socioeconomic and family environment	The earthquake reduces income, damages housing, and increases conflict at home (Conger and Donnellan, 2007; Almond, Currie, and Duque, 2018)
Transmission of maternal stress	Infants regulate their stress through the mother. If she is stressed, she passes it on through daily contact (Persson and Rossin-Slater, 2018)

Data



Survey	ELPI, Early Childhood Longitudinal Survey, Chile
Sample	9,996 adolescents born between January 2006 and August 2009, aged 6 to 49 months on 27-F; 4,961 girls
Respondents	The caregiver in every round; the child in 2017 and 2024

▸ Descriptives

Econometric specification

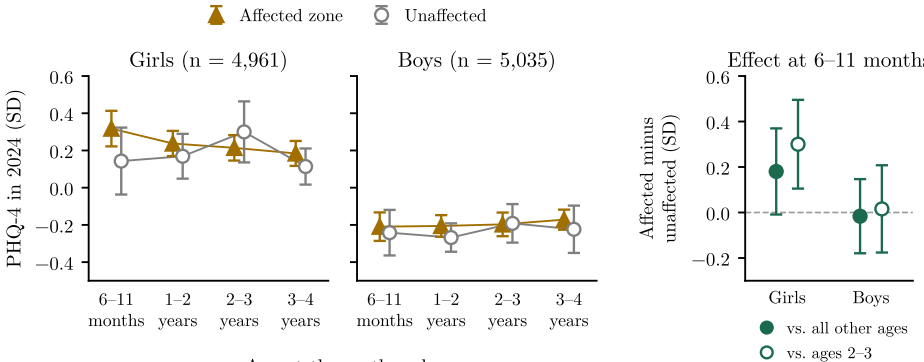
$$Y_{imc} = \beta \left(\text{Age}_i^{6\text{–}11} \times \text{Affected}_m \right) + \sum_a \gamma_a \, \text{Age}_{ai} + \psi_m + \theta_c + X_i' \delta + \varepsilon_{imc}$$

Y_{imc}	Mental health at ages 14–18, PHQ-4
$\text{Age}_i^{6\text{–}11}$	Aged 6–11 months at the earthquake; comparison, all other ages
Age_{ai}	Age groups: 6–11 months, 1–2, 2–3, and 3–4 years
Affected_m	Municipality in the zone of destructive damage
ψ_m, θ_c	Municipality and birth-year fixed effects
X_i	Linear trend in month of birth; child and household controls

Age groups in months: 6–11, 12–23, 24–35, and 36–49. The earthquake date is the same for everyone, so age at the earthquake is a function of the month of birth. Replacing γ_a, θ_c, and the trend with month-of-birth fixed effects leaves the estimates unchanged. The profile by age relative to ages 2–3 is in Table A4, panel B.

▸ Balance ▸ Attrition

Effect on mental health

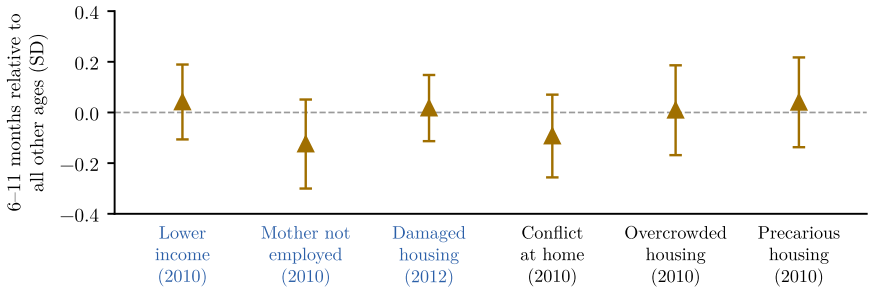


PHQ-4 (Kroenke et al., 2009), four questions answered by the adolescent on anxiety and depression. Left, the mean in each zone by age at the earthquake; in unaffected municipalities, girls aged 2–3 score unusually high. Right, affected minus unaffected for those aged 6–11 months, relative to all other ages and to ages 2–3; 95% CI. For girls, wild bootstrap $p = 0.08$; with peak ground acceleration instead of the zone, 0.19 SD in the average affected municipality ($p = 0.04$).

▸ PHQ-4 distribution ▸ Table

Socioeconomic and family environment does not explain the gap

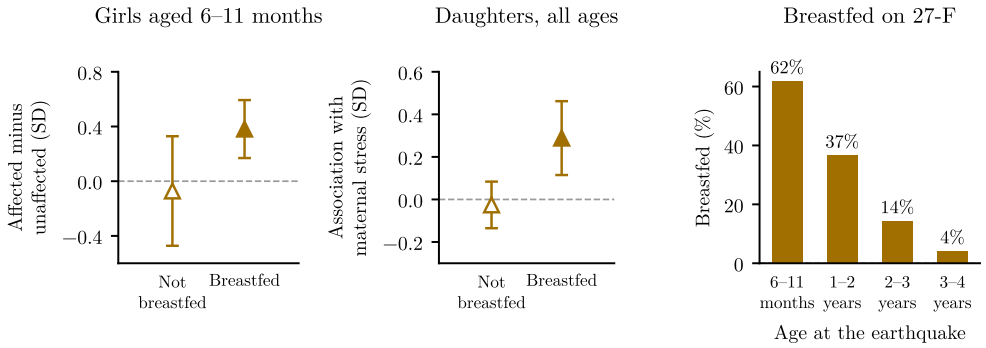
The earthquake worsened the environment (Gillmore, 2026), but not more at 6–11 months



Measured in 2010, one to nine months after the earthquake, except housing damage, reported in 2012. In blue, the mechanisms in Gillmore (2026). Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack (*mediagua*) or a lent or shared dwelling. Girls aged 6–11 months relative to all other ages; 95% CI.

▸ Table

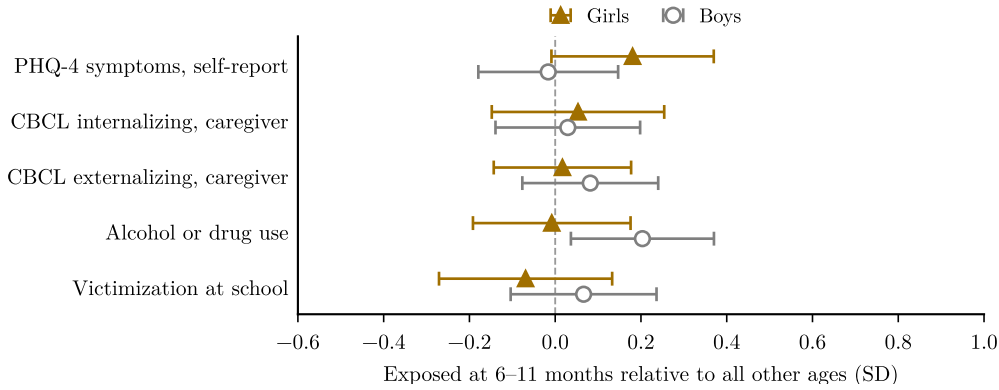
The effect at 6–11 months is concentrated among breastfed girls



Breastfed, still breastfed on 27-F (2012 report). Left, affected minus unaffected for girls aged 6–11 months, relative to all other ages; the gap between breastfed and not breastfed, 0.45 SD (SE 0.23; wild bootstrap $p = 0.07$), does not depend on the comparison group. Center, the association with maternal stress reported in 2012, which rose by 13 points in affected municipalities and accounts for about 0.01 SD of the effect.

▸ Table ▸ By age ▸ Maternal stress ▸ Means

Among boys, no effect on symptoms but more substance use



‣ Table, girls ‣ Table, boys

Findings

- Girls exposed at 6–11 months have worse mental health than those exposed at older ages, by 0.18 SD on the PHQ-4; boys show no difference.
- The socioeconomic and family environment did not worsen more for those aged 6–11 months and does not explain this difference.
- The effect at 6–11 months is concentrated among girls who were breastfed on 27-F, 62% of that age group.
- Maternal stress reported in 2012 is associated with worse mental health only among breastfed daughters, but accounts for about 0.01 SD of the effect.

Open questions

- Why do symptoms appear only among girls?

  Hypothesis: early stress leaves a latent vulnerability that surfaces in adolescence, when the gender gap in depression opens (McCrory and Viding, 2015; Salk, Hyde, and Abramson, 2017).

- What does breastfeeding transmit, if the stress the mother reports accounts for little of the effect?

  Hypothesis: the mother's acute stress response in the months after the earthquake, which a report two years later captures only in part.

References

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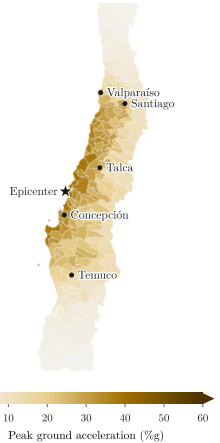
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Figure A1—Peak Ground Acceleration by Municipality



Notes: Peak ground acceleration of the 27-F earthquake, in percent of gravity, at the centroid of each municipality. Affected municipalities in the main specification are those of the six regions with destructive shaking; the map shows how shaking varied within and across them. *Source:* USGS ShakeMap of the event.

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Table A1—Descriptive Statistics

	All		Mean by sex	
	(1) Mean	(2) SD	(3) Girls	(4) Boys
Panel A. Adolescent, 2024				
PHQ-4 score (0–12)	4.32	2.68	4.89	3.78
PHQ-4 of 6 or more (share)	0.27	0.44	0.34	0.19
Age in August 2024 (years)	16.76	1.05	16.76	16.76
Age at the earthquake (months)	27.06	12.63	27.07	27.06
Affected municipality (share)	0.78	0.42	0.78	0.78
Panel B. Birth and household, 2010				
Birth weight (kg)	3.39	0.50	3.34	3.44
Gestation (weeks)	38.64	2.01	38.69	38.59
Mother's age (years)	29.52	7.11	29.60	29.44
Mother's education (years)	10.31	3.18	10.28	10.34
Household size	4.93	1.68	4.92	4.93
Rural (share)	0.09	0.29	0.09	0.09
Panel C. After the earthquake, 2012				
Breastfed on 27-F (share)	0.24	0.43	0.24	0.24
Mother reported stress (share)	0.14	0.34	0.14	0.13
Adolescents	9,996		4,961	5,035

Notes: Children of the 2010 baseline (15,175) interviewed in 2024 (10,003) with a PHQ-4 score (9,996); ELPI sampling weights. Panel C was reported in 2012; breastfed means still breastfed on 27-F.

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Table A2—Balance on Characteristics Fixed Before the Earthquake

	At birth (SD)			Mother, 2010 (years)	
	(1) Weight	(2) Length	(3) Gestation	(4) Education	(5) Age
Panel A. Relative to all other ages					
6–11 months × Affected	0.075	−0.034	0.015	−0.280	−0.240
	(0.093)	(0.078)	(0.103)	(0.279)	(0.682)
Panel B. Relative to ages 2–3					
6–11 months × Affected	0.193	0.036	0.110	−0.560	−0.129
	(0.118)	(0.096)	(0.111)	(0.333)	(0.775)
1–2 years × Affected	0.105	0.031	0.096	−0.481	−0.139
	(0.103)	(0.100)	(0.086)	(0.264)	(0.695)
3–4 years × Affected	0.245	0.172	0.182	−0.350	0.457
	(0.083)	(0.099)	(0.075)	(0.300)	(0.603)
Observations	4,530	4,525	4,818	4,882	4,920
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No

Notes: Girls. Each column regresses a characteristic fixed before the earthquake on age group × affected, with municipality, birth-year, and age-group fixed effects and no controls; panels as in Table A4. Birth weight, length, and gestation are standardized; the mother's education and age were reported in 2010. Among the 4,190 girls with birth records, the PHQ-4 coefficient for those exposed at 6–11 months, relative to all other ages, is 0.138 (0.093) without and 0.139 (0.093) with the three birth outcomes as controls. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

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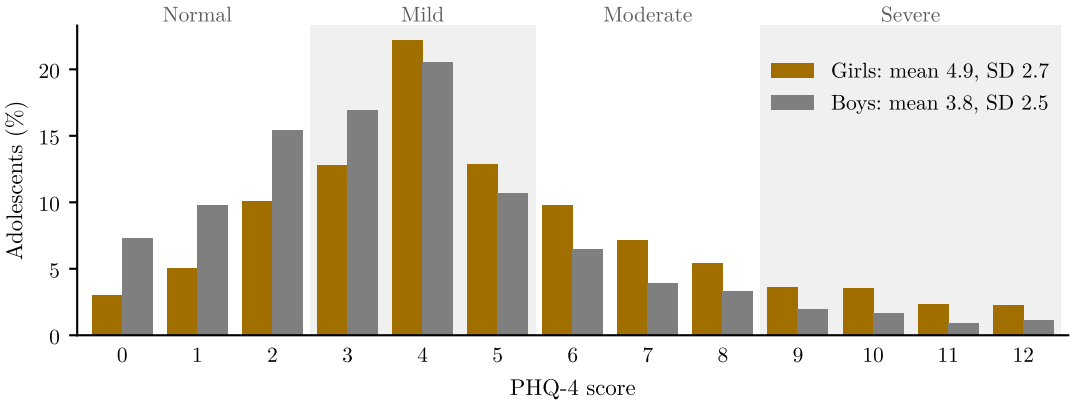
Table A3—Attrition between 2010 and 2024

	Interviewed in 2024			Birth weight, girls (SD)	
	(1)	(2)	(3)	(4)	(5)
	Girls	Boys	All	2010 baseline	Interviewed in 2024
Panel A. Relative to all other ages					
6–11 months × Affected	−0.057	−0.007	−0.034	−0.015	0.031
	(0.038)	(0.037)	(0.027)	(0.087)	(0.103)
Panel B. Relative to ages 2–3					
6–11 months × Affected	−0.047	0.019	−0.019	0.064	0.120
	(0.044)	(0.043)	(0.030)	(0.098)	(0.115)
1–2 years × Affected	0.009	0.049	0.026	0.087	0.073
	(0.035)	(0.035)	(0.025)	(0.075)	(0.092)
3–4 years × Affected	0.019	0.028	0.019	0.145	0.189
	(0.035)	(0.034)	(0.024)	(0.076)	(0.093)
Joint test, panel B, *p*-value	0.47	0.57	0.43	0.30	0.24
Mean of dependent variable	0.67	0.67	0.67		
Observations	6,990	7,171	14,161	6,409	4,300
Region FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes

Notes: Children with the 2010 developmental assessment, which gives age in months (14,161 of 15,175); panels as in Table A4. Columns 1–3 equal one if interviewed in 2024; columns 4–5, standardized birth weight of all girls assessed in 2010 and of those interviewed in 2024. Age at the earthquake is the age at the 2010 assessment minus the months from 27-F to the region's median assessment month, for stayers and leavers alike. The 2010 files do not identify the municipality, so region fixed effects, no weights, and robust standard errors.

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Figure A2—Distribution of the PHQ-4 Score



Notes: PHQ-4 score in 2024, weighted with ELPI sampling weights. The score adds four items scored 0 to 3; Kroenke et al. (2009) classify 0–2 as normal, 3–5 as mild, 6–8 as moderate, and 9–12 as severe.

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Table A4—Effect on Mental Health

	PHQ-4 (SD)				
	(1) All	(2) Girls	(3) Boys	(4) Girls	(5) Girls
Panel A. Relative to all other ages					
6–11 months × Affected	0.077	0.180	−0.016	0.190	0.066
	(0.057)	(0.097)	(0.083)	(0.097)	(0.030)
Panel B. Relative to ages 2–3					
6–11 months × Affected	0.147	0.300	0.016	0.310	0.096
	(0.069)	(0.100)	(0.098)	(0.101)	(0.029)
1–2 years × Affected	0.102	0.171	0.052	0.178	0.054
	(0.070)	(0.098)	(0.085)	(0.099)	(0.031)
3–4 years × Affected	0.103	0.182	0.042	0.177	0.034
	(0.060)	(0.086)	(0.086)	(0.089)	(0.028)
Observations	9,996	4,961	5,035	4,961	4,961
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	No	Yes
Age-group FE	Yes	Yes	Yes	No	Yes
Month-of-birth FE	No	No	No	Yes	No
Exposure	Zone	Zone	Zone	Zone	PGA
Controls	Yes	Yes	Yes	Yes	Yes
Wild bootstrap p-value, panel A	0.185	0.075	0.866	0.068	0.043

Notes: Affected minus unaffected municipalities in the standardized PHQ-4; panel A compares ages 6–11 months with all other ages, panel B each age group with ages 2–3. Controls are a linear trend in month of birth, sex, the mother's age and education, household size, rural residence (2010), the mother's number of children, and family history of mental illness (2012). Column 4 uses month-of-birth fixed effects; column 5, peak ground acceleration per 10 points of percent of g (28 more in the average affected municipality). Wild bootstrap with 9,999 Rademacher draws under the null. ELPI weights. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

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Table A5—The Socioeconomic and Family Environment

	Mechanisms in Gillmore (2026), SD			Other household conditions, SD		
	(1) Lower income	(2) Mother not employed	(3) Damaged housing	(4) Conflict at home	(5) Overcrowded housing	(6) Precarious housing
Panel A. Relative to all other ages						
6–11 months × Affected	0.041	−0.125	0.017	−0.093	0.009	0.040
	(0.075)	(0.090)	(0.067)	(0.083)	(0.090)	(0.090)
Panel B. Relative to ages 2–3						
6–11 months × Affected	0.056	−0.167	0.044	−0.143	0.114	0.094
	(0.084)	(0.103)	(0.069)	(0.096)	(0.099)	(0.095)
1–2 years × Affected	0.024	−0.116	0.013	−0.154	0.121	0.048
	(0.081)	(0.092)	(0.051)	(0.084)	(0.100)	(0.083)
3–4 years × Affected	0.021	−0.011	0.063	0.002	0.188	0.108
	(0.086)	(0.075)	(0.054)	(0.062)	(0.087)	(0.081)
Observations	4,478	4,925	4,423	4,701	4,961	4,960
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Girls. Each column is a standardized index of the household environment; higher is worse. Measured in 2010, one to nine months after the earthquake, except housing damage, reported in 2012. Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack or a lent or shared dwelling. Panels and specification as in Table A4. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

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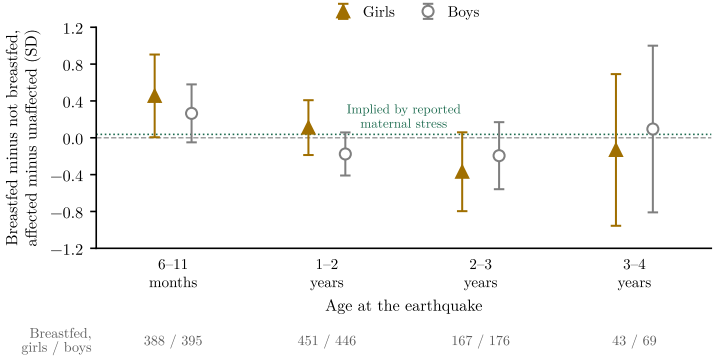
Table A6—Breastfeeding and the Effect at 6–11 Months

	PHQ-4 (SD)		
	(1) Girls	(2) Girls	(3) Boys
6–11 months × Affected × breastfed	0.381	0.362	0.125
	(0.108)	(0.110)	(0.102)
6–11 months × Affected × not breastfed	−0.072	−0.074	−0.140
	(0.204)	(0.210)	(0.127)
Difference, breastfed − not breastfed	0.453	0.436	0.265
	(0.229)	(0.232)	(0.162)
Wild bootstrap p-value, difference	0.074	0.095	0.115
Observations	4,409	4,338	4,465
Municipality FE	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes
Age group × breastfed FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Mother's education and employment × breastfed and × affected	No	Yes	No

Notes: Difference between affected and unaffected municipalities in the standardized PHQ-4 for those aged 6–11 months, relative to all other ages, separately for those still breastfed on 27-F (2012 report) and those who were not. Column 2 lets breastfeeding and the affected zone vary with the mother's education and employment (2010). Controls, weights, and wild bootstrap as in Table A4. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

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Figure A3—Breastfeeding and the Effect of the Earthquake by Age



Notes: For each age at the earthquake, the difference between affected and unaffected municipalities among those breastfed on 27-F minus the same difference among those who were not, with fixed effects for age group × breastfed, municipality, and birth year, and the controls of Table A4. The dotted line is the value implied by maternal stress reported in 2012 (0.29 × 0.13). 95% CI.

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Table A7—Maternal Stress, Breastfeeding, and Daughters' Mental Health

	Daughters' PHQ-4 (SD)		
	(1) Baseline	(2) Mother's SES	(3) Mother's SES and infancy
Maternal stress × not breastfed	−0.026	−0.005	−0.004
	(0.056)	(0.063)	(0.064)
Maternal stress × breastfed	0.288	0.280	0.286
	(0.089)	(0.096)	(0.112)
Difference, breastfed − not breastfed	0.314	0.285	0.290
	(0.108)	(0.102)	(0.110)
Observations	4,409	4,377	4,377
Municipality FE	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes
Age group × breastfed FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Stress and breastfed × mother's education and employment	No	Yes	Yes
Stress × 6–11 months	No	No	Yes

Notes: Girls. Maternal stress equals one if the mother reported stress, distress, or anxiety after the earthquake (2012); breastfed equals one if the child was still breastfed on 27-F (2012 report). Column 1 is the estimate in the figure; column 2 lets stress and breastfeeding vary with the mother's education and employment (2010); column 3 also lets stress vary at 6–11 months. Controls and weights as in Table A4. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

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Table A8—Maternal Stress and Breastfeeding by Age at the Earthquake

	Mothers with stress (%)		Breastfed (%)
	(1) Affected	(2) Unaffected	(3) On 27-F
6–11 months	15.0	3.8	61.8
	(1.2)	(1.1)	
1–2 years	15.6	3.6	36.6
	(0.8)	(0.8)	
2–3 years	17.8	3.7	14.3
	(0.9)	(0.8)	
3–4 years	17.8	3.2	4.1
	(0.8)	(0.7)	
Observations	6,772	2,132	8,874

Notes: Unweighted means, with standard errors in parentheses, by the child's age at the earthquake, among the adolescents of Table A1 whose mother answered each question in 2012. Columns 1 and 2 report the share of mothers who reported stress, distress, or anxiety after the earthquake, in affected and unaffected municipalities; column 3, the share of children still breastfed on 27-F.

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Table A9—Other Outcomes, Girls

	Adolescent report (SD)			Caregiver report, CBCL (SD)	
	(1) PHQ-4	(2) Substance use	(3) Victimization	(4) Internalizing	(5) Externalizing
Panel A. Relative to all other ages					
6–11 months × Affected	0.180	−0.008	−0.069	0.053	0.017
	(0.097)	(0.094)	(0.103)	(0.103)	(0.082)
Panel B. Relative to ages 2–3					
6–11 months × Affected	0.300	−0.032	−0.030	0.119	0.024
	(0.100)	(0.109)	(0.114)	(0.106)	(0.087)
1–2 years × Affected	0.171	0.026	0.044	0.154	0.100
	(0.098)	(0.096)	(0.081)	(0.120)	(0.107)
3–4 years × Affected	0.182	−0.094	0.081	0.042	−0.082
	(0.086)	(0.083)	(0.092)	(0.095)	(0.083)
Observations	4,961	4,961	4,539	4,748	4,748
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Girls. Each column is a separate regression with the specification and panels of Table A4. The adolescent reports the PHQ-4, substance use (any alcohol, tobacco, cannabis, or other drug in the last 12 months), and peer victimization and class climate; the caregiver reports internalizing and externalizing problems (CBCL). All measured in 2024 and standardized; higher is worse. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

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Table A10—Other Outcomes, Boys

	Adolescent report (SD)			Caregiver report, CBCL (SD)	
	(1) PHQ-4	(2) Substance use	(3) Victimization	(4) Internalizing	(5) Externalizing
Panel A. Relative to all other ages					
6–11 months × Affected	−0.016	0.203	0.066	0.029	0.082
	(0.083)	(0.085)	(0.087)	(0.086)	(0.081)
Panel B. Relative to ages 2–3					
6–11 months × Affected	0.016	0.172	0.136	0.012	0.144
	(0.098)	(0.104)	(0.106)	(0.100)	(0.079)
1–2 years × Affected	0.052	−0.064	0.112	−0.063	0.053
	(0.085)	(0.089)	(0.086)	(0.099)	(0.092)
3–4 years × Affected	0.042	−0.031	0.098	0.010	0.126
	(0.086)	(0.079)	(0.094)	(0.099)	(0.090)
Observations	5,035	5,034	4,637	4,804	4,804
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Boys. Outcomes and specification as in Table A9. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

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